

Pawel Renc

Healthcare AI / Foundation Models / ML Systems

Boston, MA · rencpawe@gmail.com · GitHub · LinkedIn · Google Scholar

Research profile

Research fellow at Massachusetts General Hospital working on generative models of longitudinal electronic health records and the infrastructure for training and evaluating them. Co-developer of ETHOS and ARES, combining healthcare-AI research with industry experience at Snowflake and high-performance computing research at ETH Zürich.

Research & industry experience

Research Fellow

Massachusetts General Hospital · Sitek Lab

May 2026–present

Boston, MA

- Develop EHR foundation models and end-to-end infrastructure for large-scale data processing, training and inference.
- Maintain the ETHOS research codebase, spanning tokenization, model training, evaluation and synthetic timeline generation.

Research Scientist

Snowflake · AI Research

Oct 2025–May 2026

Bellevue, WA

- Designed and developed an agentic medical-coding system for assigning diagnoses from clinical notes.
- Partnered with forward-deployed engineering teams on healthcare applications of medical foundation models.

AI Research Intern

Snowflake

Apr–Oct 2025

Bellevue, WA

- Developed and fine-tuned EHR foundation models; built training and inference workflows with Transformers, vLLM and Ray.

Visiting PhD Researcher

Massachusetts General Hospital · Harvard Medical School

Mar 2023–Apr 2025

Boston, MA

- Co-developed ETHOS for generative patient-timeline modeling and zero-shot health prediction; contributed to timeline representation, tokenization, transformer training and inference.
- Led software and formal analysis for ARES, extending timeline simulation to adaptive risk estimation.

Visiting PhD Researcher

ETH Zürich · Scalable Parallel Computing Laboratory

2022–2023

Zürich, Switzerland

- Researched GPU execution of graph neural networks and global tensor formulations; coauthored work published at SC 2023.

PhD Researcher

Sano Centre for Computational Medicine

Jan 2022–Feb 2023

Kraków, Poland

- Developed machine-learning methods for disease-phenotype discovery through patient-data clustering, in collaboration with University of Glasgow clinicians.

Technical expertise

Modeling: EHR foundation models, generative AI, transformers, graph neural networks.

ML systems: PyTorch, Transformers, vLLM, Ray; distributed training and inference.

Engineering: Python, C/C++, CUDA, Julia, Bash; Polars, SQL, Linux and GPU computing.

Selected publications

Foundation model of electronic medical records for adaptive risk estimation

First author · *GigaScience*, 2025. DOI: [10.1093/gigascience/giaf107](https://doi.org/10.1093/gigascience/giaf107)

Zero shot health trajectory prediction using transformer

First author · *npj Digital Medicine*, 2024. DOI: [10.1038/s41746-024-01235-0](https://doi.org/10.1038/s41746-024-01235-0)

High-Performance and Programmable Attentional Graph Neural Networks with Global Tensor Formulations

Coauthor · *SC*, 2023. DOI: [10.1145/3581784.3607067](https://doi.org/10.1145/3581784.3607067)

Towards efficient GPGPU Cellular Automata model implementation using persistent active cells

Coauthor · *Journal of Computational Science*, 2022. DOI: [10.1016/j.jocs.2021.101538](https://doi.org/10.1016/j.jocs.2021.101538)

EBIC.JL — an efficient implementation of evolutionary biclustering algorithm in Julia

Coauthor · *GECCO Companion*, 2021. DOI: [10.1145/3449726.3463197](https://doi.org/10.1145/3449726.3463197)

Research software

ETHOS / ARES — EHR tokenization, transformer training and health-trajectory inference. Research software supporting generative patient-timeline modeling.

EBIC.jl — GPU-enabled evolutionary biclustering in Julia; developed during master's research.

Education

Doctorate · Engineering and Technology

May 2026

AGH University of Krakow · Discipline: Information and Communication Technology

Dissertation: *Foundation model for electronic health records*.

Supervisors: Jarosław Wąs and Arkadiusz Sitek. Degree conferred May 21, 2026.

Master's degree · Computer Science

2021

AGH University of Science and Technology · System modelling and data analysis

Bachelor's degree · Computer Science

2020

AGH University of Science and Technology

Selected service & teaching

Peer review: IEEE Journal of Biomedical and Health Informatics (two rounds, 2025); IEEE Transactions on Medical Imaging (2023); BMJ Digital Health & AI (2026).

Grand rounds: Applications of Machine Learning in Healthcare, DHR Health, April 2024.

Research and Teaching Assistant, AGH

2020–2021

Taught Unix, Operations Research and Discrete Systems Modeling.

Earlier engineering experience

QED Software · Data Scientist

2020–2021

Built ML methods for detecting erroneous table joins and distinguishing bots from human game players.

Intel · Software Engineering Intern

2019

Developed Python automation for integration, unit and system-level testing.

GrowMagick · Software Engineer

2019–2020

Built data pipelines and anomaly-detection workflows for an advertising-optimization R&D project.